To Blockchain or Not To Blockchain? An E-Voting Question

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***Abstract*—When blockchains appeared with its anonymous, tamper proof and immutable features, it became a tempting option for e-voting systems and countries welcomed many pilot studies around the year 2018. Then came a batch of opposing articles of which the most famous is an MIT paper in 2021 authored by a group well-known security pioneers, joined by some blockchain experts, that describes the idea as "*Going from Bad to Worse*"; where they do not recommend internet voting, calling it bad to begin with and believe deploying blockchains will make it worse. Although most countries participated in the pilot studies eventually did not integrate blockchains in their systems, many others like Russia, Mexico, UAE, some Canada municipalities and some USA states (although debatable) did one way or another. As of today in 2026, literature holds thousands of papers proposing, or surveying, blockchain backed e-voting protocols/systems; many of which contain mixed information about deploying blockchains in the e-governmental system and using it in e-voting, about using it in private entities elections and using it in country level election, and about pilot studies or plans that was dropped at the end due to highly promoted media coverage at the beginning. In addition, to our knowledge, none of those papers defeated or discussed the arguments in the MIT paper; most of its citations came from blockchain related research or political legal research. Motivated by our own confusion as authors, this paper aims to fold the available material in four directions; 1) the opposing material with its argument, 2) the approving material with its defense, 3) the successful real world use cases at the governmental election level (i.e. separating from and/or resolving any mix-up with pilot studies and/or small/medium scale existing systems (companies/institutions/organizations) in the same country, 4) the intersection areas going back & forth between the two fields. This includes how e-voting systems benefited from the continuous evolvement of Zero Knowledge Proofs (ZKPs) research accompanied different advancements in blockchain; also, how blockchain has innovated new vote buying/coercion techniques that take advantage of the anonymous automated online coding feature or the immutable public ledger feature. Finally, before concluding the paper, we investigate two existing blockchain-based e-voting systems and analyze them in light of the existing criticism and in comparison with similar ones if existed.**

Keywords—e-voting, blockchains, Zero Knowledge Proofs (ZKPs), vote coercion, vote buying, Decentralized Autonomous Organizations (DAOs).

I. Introduction

A simple search query on the two keywords "*e-voting*" and "*blockchain*" will lead to 4725 papers + 1 Book in Springer Nature, 3960 in ScienceDirect, 673 papers + 6 Books in IEEE Xplore and 25 in MDPI; this contains more than 200 review articles as can be deduced from Table I. Although the rapidly increasing numbers, nearly 1600 papers up till now in 2026, they are not all directed to political country level voting systems. Especially during COVID-19 pandemic, online voting (with or without blockchains) became popular on all scales; hence, a lot of research articles proposed systems that are only applicable to small/medium scale cooperations (ex. [1/Table8]) which expands the scope of the mentioned numbers beyond the country-level case. Checking case-by-case in the two smallest numbers (25 and 57), only 18 papers explicitly propose a nationwide system or a system/framework that can be applied at a large-scale country level. In earlier years, developed countries [2-5] dominate studies or proposals for a specific country, while many proposals were suggested for developing countries in the last few years [6-15]. As for review articles through the years, [4] is an early 2018 example, [16] is an 2022 example that discusses the later inclusion of sensor networks (IoTs) to capture physical data (whether fingerprint/biometric authentication or touch screen inputs at poll stations); finally [17] in 2026 tries to set a comprehensive framework when the field has matured enough.

TABLE I. SEARCH ON "E-VOTING BLOCKCHAIN"

Publication	***Total***	***2026***	***Review (2026)***	***Book***
Springer	4725	773	158 (40)	1
ScienceDirect	3960	735	515 (117)	-
IEEE Xplore	673	57	>9 (7)	6
MDPI	25	4	5	-
Totals			**>687 (164)**	**7**

In general, there are two main approaches, [5], for deploying blockchains into e-voting systems. The first is to use a well-known public permissionless blockchain (ex. Ethereum [4/sec. 3.3, 5,7]) as an immutable anonymous public record with either a single vote or a batch of votes forming a transaction [18]; variants include hashes of votes, Non-Interactive Zero Knowledge Proofs (NIZKPs) or Succinct Non-interactive Arguments of Knowledge (SNARKs) as well. The second is to replace the usual Public Bulletin Board (PBB), [3/page 50], where election results are published at the end, usually by a private permissioned blockchain owned by election stake holders [4/sec.3.1&3.2, 16] or could be public in some deigns though [5]; in that direction, researchers have proposed different consensus mechanisms (the validation criteria and the way in which participating nodes negotiate their decision) , like *Proof of Partition* [19] or *Proof of Authority* [13]. It is important to note that some designs deploy both private and public blockchains; for example Agora design since 2018 [20], Fig.1, that uses a private permissioned *Skipchain* [20/ref. 52] to provide a faster consensus for ballots, then stores snapshot hashes of *skipchain* on the Bitcoin

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blockchain as a non-standard UTXO[1] (*Catena* schema [20/ref. 58]). In 2024/2025 newer designs (ex. *Vocdoni* [21] in Fig.2) used Ethereum in a similar manner where ballots are stored in a custom Zero Knowledge layer2 chain (L2 ZK-rollup), then batched cryptographically into Ethereum base layer L1; *opn.vote* [22], Fig.3, on the other hand used *Inter Planetary File System (IPFS)* (a peer-to-peer decentralized storage and file sharing protocol[2]) for actual ballot storage, while only meta data are stored inside the Voting Smart Contract (VSC) on Ethereum.

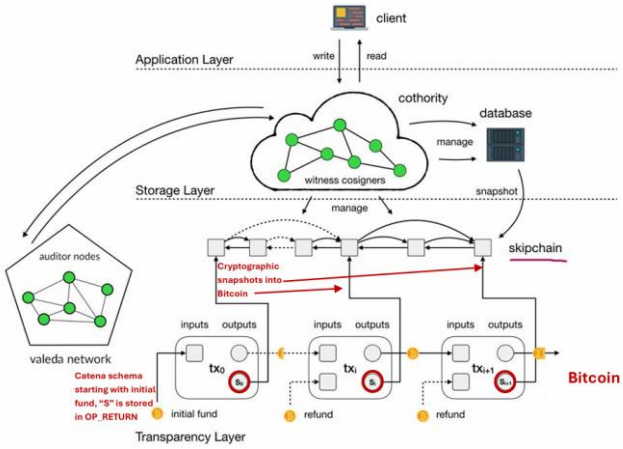


Fig. 1. Agora blockdiagram taken from [20] with some clarifying annotations added in red.

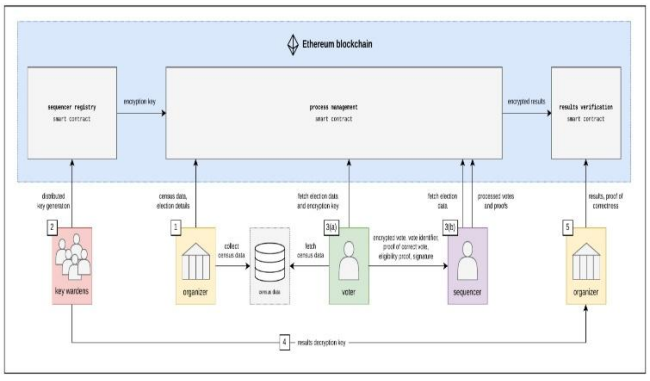


Fig. 2. Vocdoni protocol flow diagram taken from [21].

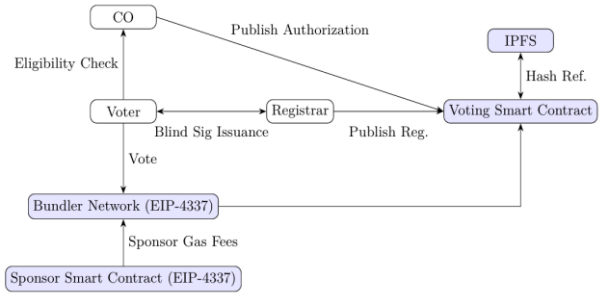


Fig. 3. Opn.vote high level architecture taken from [22].

Also, the reader should be aware that most blockchain voting design approaches can be suggested for remote online (internet) voting or poll station electronic voting using different kinds of voting machines. Lastly, the research direction discussing/testing public trust and/or acceptance to such systems, like [23-25], is beyond the scope of this paper.

Due to the huge existing material, this paper will follow a time chronological order; section II will summarize important pilot studies in the early years 2016-2018 and whether aborted or adopted, then section III will discuss the opposing research papers started appearing from 2018 up till 2021. Section IV will follow with other interactions between the two fields like new attacks that that became possible with blockchains. Then, section V will discuss two case studies of officially used blockchains based e-voting systems (zk-voting in South Korea and Voatz with all the debates around it), while section VI concludes the paper. Finally, this paper assumes familiarity with basic blockchain and e-voting terminology; unfamiliar readers may find a preface and related terminology file in [].

II. Experiments & Pilot Studies

Researchers in e-voting have long recognized a tradeoff between privacy (coercers cannot link counted votes to their true voters) and verifiability (voters should be able to verify the correct count of their votes to guarantee the integrity of the election). Blockchains seem to provide an easy way to achieve both; especially with the online coding capabilities added by Ethereum in 2015. Hence, many countries around the world started small-scale pilot studies, or funded research to test the water on the idea. This section summarizes important pilot studies and tracks the current status of their voting systems.

A. European Countries

In **Ukraine**, *e-Vox*, an NGO of developers who won of the first Ukrainian contest of IT-projects on electronic democracy [26], developed and piloted probably the first blockchain e-voting system (on Ethereum) for city hall elections in Kiev since 2016 [27, 3/sec.4.2.3] and kept seeking to generalize the experiment up till 2021 [28]. **Switzerland**, with *Luxoft* company, performed an online voting experiment using blockchains (Hyperledger Fabric; private permissioned chain by IBM) on the city of Zug in 2018 [4/sec.3.1, 29,30]; 72 citizens out of 220 holding digital identities voted on things like whether fireworks should be allowed on festivals or not. Also in 2018, *Smartmatic* company offered to **Norway** a trial blockchain based online voting, *TIVI* [31,32], on a local referendum. **Estonia**, on the other hand investigated then rejected [33,34] proposals to use blockchains in their long-standing internet voting system (since 2005, but the new design *IVXV*, which Smartmatic is part of, was first used in 2017).

Currently, blockchains maybe used by the e-government in some European countries (ex. Estonia [35,36]); however, they were only officially used as *a voting reporting tool* by **Romania** [37] in its 2023 national parliamentary elections. Although countries like Germany [38,39,40] and Norway are

[1] Since Bitcoin allows only primitive scripts for coin transfer (does not have a smart contract capability), people used to piggyback their applications/chains to Bitcoin through what is called non-standard UTXOs; they write what seems to be a coin transfer TX, then put the value they care to post (usually cryptographic hash) in the OP_RETURN field. Catena schema starts with a Bitcoin value *n* stored in a UTXO address *A* and each TX transfers a small amount from *A* to another UTXO; this can go on till *A* spends all the *N* value. Then, *Catena* in Agora design is another custom blockchain, managed by those who have consensus rights on the election, where each block contains the value and link to one of those TXs; this way Bitcoin can expose any manipulations of those values.

[2] https://ipfs.tech/; the paper behind it (https://arxiv.org/abs/1407.3561).

still skeptical about the integrity of internet voting in general, there was a successful Ethereum-based pilot *opn.vote* that was deployed in **Germany**'s referendum in Sep 2025; two papers were dedicated to the experiment [22,41].

On the non-governmental level, some political parties use blockchain e-voting in Estonia and Spain [37] while the most famous company working in Germany, *Polyas*, does not use blockchains [42] (at least before the success of *opn.vote*). However, there is a success story for *Vocdoni*, it developed a (Zero Knowledge Succinct Non-Interactive ARgument of Knowledge) zkSNARK-based state machine L2 on Ethereum [21], with the opposition in **Belarus** [43] and with Omnium Cultural, a non-governmental organization in **Barcelona,** with over 180,000 members [44].

B. Russia

In Sep 2019, a pilot project conducted e-voting using a private blockchain "*Sovereign*" for the Moscow City Duma in 3 electoral districts [18,45], increased to 6 in 2021[46]. The system was first tested on a university students council voting in July along with an international competition to hack it organized by the Civic Chamber whom Pierre Gaudry (Inria, France) won its half; [18,47] detail the system architecture, the earlier attacks and how they were handled. Putin has officially signed the regulations for holding online voting for all elections and referenda in March 2022 [48], and Russia remained to expand the usage of Distributed Ledger Technology (DLT) in its internet voting system (known in Russian language as DEG [49.50]) through private permissioned blockchains, in spite the criticism [46,51]. However, after the Sep 2025 elections, Russia declared a temporary rollback [52] of internet voting during the war.

C. Britain (UK)

Proposals in research articles can be traced back to 2018 [4,53]. A pilot experiment was conducted in April 2019 by a team from London School of Economics [54]; they used, [55], the open source of "*Verify my Vote*" with the voting system owned by Electoral Reform Services. Although UK government still funds flexible voting pilots [56], and research continues to further investigate the subject [57,58], online blockchain voting is still viewed as high-risk, [59], compared to traditional paper ballots.

D. Australia

Conducted some pilot studies around with a lot of promoting publicity, [60], but currently does not deploy internet voting in its nationwide elections [61/electronic voting]. To be accurate, sometimes voters have the option to vote electronically [62/alternative voting methods] in designated booths using Kiosk touch screens; the counting software, *eVACS* [63], is written in Ada programming language and does not use blockchains (available as opensource).

E. Africa: Sierra Leone

A Swiss company, Agora [20], ran a pilot project in the Mar 2018 presidential election that was reported in the literature as a proof of concept for the value blockchain e-voting [64,65]; Electoral staff manually recorded all votes of the western District into *Agora*'s blockchain platform to demonstrate its potential for transparency and fraud prevention. However, the company promoting it as the first ever blockchain elections [66] and also rushing to publish its results before the official ones [64], led to the government issuing a public denial, [67,68], clarifying that blockchain was not used in the official vote count and that *Agora* had no formal role in the election process.

F. Asia

In Sep 2018, **Japan** held a trial blockchain-based voting (for social contribution projects) in Tsukuba city [69], while **Thailand** main opposition party deployed the *Zcoin* blockchain (a privacy focused cryptocurrency that preceded *Zcash* and rebranded to *Firo*.org in 2020)[3] in its Nov 2018 election with more than 120,000 members [70]. Although [4/sec.3.2], 2018, mentions a pilot in Bandung **Indonesia** using 2 private machines running the Ethereum protocol, we could not trace another credible source on that; [71] which is written by researchers from Bandung and published in an Indonesian journal did not mention such a thing in its survey part. Finally, **South Korea** started its first blockchain voting trial, [72], in Dec 2018 based on Hyperledger Fabric (private permissioned blockchain designed by IBM). Currently, we traced two important online voting companies in the country; *kevoting*, [73], not based on blockchains, used in **Kyrgyzstan** public office elections 2023 [73/2023 (04) & press (15,16)], and was about to be used in their 2026 parliament elections 2026[4] but plans were cancelled [74]. On the other hand, *zkrypto* whose product *zkvoting* deploys blockchains and is officially used by the ***National Election Commission (NEC)***, [75], for public institutions elections and local community referenda [75/video]; however, the system is not used for public elections, [76], where poll station voting machines are used. Their system will be thoroughly discussed is section IV.

G. Brazil

Brazil depends solely on poll stations electronic voting machines since 1996 up till now, with some criticism by security experts [77,78]. In 2020, the country requested proposals from technology companies to conduct a trial blockchain-based online voting with a sample of voters from three cities choosing fictitious candidates [79,80]. Also, in May 2025, a senator requested [81] to explicitly include blockchains and distributed ledgers among the technologies that may be considered for the purposes of studies and the development of new voting solutions.

H. Guatemala

In the 2nd round of 2023 presidential elections some Bitcoin developers from Guatemala civil society [82], in conjunction with Peter Todd a core Bitcoin developer, has done something partially similar to Cotena chain [20/sec.3.1.2] in Agora design (footnote 1). Peter Todd's *Open Timestamp* protocol [83] calculates a cryptographic hash of the submitted

[3] Zcash started 2014 as an improvement on Zcoin to hide also the transaction destination (would be the Election Authority EA in e-voting) and the payment amount; this will be further discussed in detail in section 4 on nullifiers. For now, the interested reader can check this 2017 article (https://medium.com/the-zcoin-digest/zcoin-compared-to-zcash-4ad21d97636f); in 2025 (https://phemex.com/academy/what-is-firo-firo-guide).

[4] This link (https://qazinform.com/news/online-voting-planned-for-kyrgyzstans-next-parliamentary-elections-a82307) precedes the cited one in [74] by 2 weeks.

data along with the timestamp it was submitted to the protocol, then posts it to the Bitcoin blockchain; during the election, [84], timestamped fingerprints of partial results (say a Ballots box along with the timestamp it was created) were posted on Bitcoin. The process was promoted as "***Immutable Democracy***" [85], since Bitcoin was used as an immutable cryptographic witness on each Ballot box at a certain point of time against doubts of manipulations or altering in the aggregation process.

I. USA

As a first world country with the most advanced technologies, one of the earliest blockchain e-voting pilots, March 2016 [86], was conducted by Smartmatic company; they ran a small trial in the **Utah** branch of the republican party voting for the presidential candidates; only usability issues were reported by voters. Seeming to be a supporting state for the technology, the first official blockchain vote (a semi-private permissioned Bulletin Board) was casted from Utah (using *Voatz*) in the October 2020 presidential elections [86]; the vote went to an independent cryptocurrency entrepreneur candidate [87].

Between the two incidents, 2018/2019, there were more than 30 blockchain-based e-voting pilots by *Voatz* [88] at the country level in three other states; West Virginia, Colorado, and Oregon. A 2.2m$ funded pilot experiment took place in **West Virginia** primary elections 2018; the pilot targeted overseas military members, [89], where only 2 sample offices with couple of dozens of voters participated. Despite an attempted hack to West Virgina pilot [90], and wide concerns raised by academia on blockchain e-voting [86, 91-93], *Voatz* pilots were repeated for overseas citizens in **Colorado** Denver, Mar 2019 [88,94], then in **Oregon** two counties Jackson and Umatilla, Sep 2019 [95]. Through the years, researchers from MIT [96], and other places [97,98], got in a long battle against Voatz, but the fact is it is still used, [99], in many USA states and **Canada** municipalities and the **Mexican** elections as their official site, [100], acknowledges.

The too many rejecting voices of internet voting in general[5] [101-106], and blockchain based systems in specific [86, 91-93, 107-112], prevented allowing it as a general option for country residents in USA. However, there are two groups of special case citizens whom their vote privacy will be jeopardized otherwise; without *Voatz* or a similar app, they have to sign a waiver of their vote privacy rights when sending their votes by regular post office mail [89, 94]; this holds longer delays to the election results as well. The first group that was included in almost all pilots is overseas citizens, especially military personnel (and their families) who are abroad in duty call; they vote according to Under the federal ***Uniformed and Overseas Citizens Absentee Voting Act (UOCAVA)***, [113], and the *Defense Advanced Research Projects Agency (DARPA)* is continuously sponsoring research for more secure options[6]. The second group is people with disabilities who state governments first faced some resistance when expanding *Voatz* usage to them in West Virginia [114], but then was included. As of now, Aug 2026, the law dictates a digital alternative (could be mail, Fax, or a web portal like Voatz) for those two groups, [115/interntvoting/], in 32 states and Washington DC, while internet voting is completely banned in 18 states (ex.: Georgia, Ohio, Illinois, Maryland, Minnesota, New Hampshire, New York, Wisconsin).

However, some steps has been taken recently in the banning states; follows is some examples that targeted the general election population. In April 2025, **New York** Assembly member Clyde Vanel (chair of subcommittee on Internet and New Technology) reintroduced a newer version of his proposed *Blockchain for Election Integrity Act* [116,117], which he tried to pass four times before; the bill, [118] proposed to conduct a formal thorough study by experts on how blockchain could help secure voter records and improve transparency. According to the official NY senate site [118], the bill is still active and has moved through the proper channels reaching the state senate on Jan 2026.

In **Georgia** [119] on the other hand, blockchain technology paved its way to Nov 2024 presidential elections; Screven county applied the same technique that was applied in Guatemala through *Simple Proof*, [120], (an election specific startup founded by the same Peter Todd in [83]) to post a timestamped cryptographic proof of the poll station election results (as taken from the air gapped voting device in the poll station) on the Bitcoin blockchain. In this technique, blockchain technology just adds an extra security layer without changing the existing systems or disrupting their functionality. Although banning internet voting in general [115], this was more than justified for Georgia after previous problems, [121], that happened in 2020 elections regarding some partial results (certain voting machines); other similar problems [122], for example the Wisconsin 2024 recount (due to suspected tampering of records from a certain machine for leaving its door open during night), could have been replaced by calculating the cryptographic fingerprint of the current data and comparing it to the one posted on Bitcoin[7]. Note however, that this technique does not prove the integrity of the results before the time it was hashed; it could have a reverse effect of asserting fraud data if data was somehow manipulated before the proof is generated. Josh Benaloh described it in [86] as taking a picture of an existing picture, we say yes it is not like a picture from the actual seen, but it is much better as a proof. If a losing candidate took a photocopy or a mobile photo of the subtotals counting sheets, the Election Authority (EA) and/or the winning candidate could claim in court that the photos are fake and the court has no way of knowing (beyond reasonable doubt) which sheets are fake; while, if the cryptographic fingerprint of the sheets is on a public trusted blockchain, we could think of the blockchain as a trusted as an analogy to a "never-lie & never-forget" hear saying witness

[5] According to [92], a famous quote of Ronald Rivest (one of the designers of RSA) is "*Best practices for internet voting are like best practices for drunk driving- there is no safe way to do either*".

[6] Read this preprint (https://arxiv.org/abs/2411.11796) rejecting a DARPA funded protocol proposed for remote voting, then read the acknowledgement in the updated version of the same MERGE protocol (https://link.springer.com/chapter/10.1007/978-3-032-05036-6_1) thanking the authors for their criticism that led to significant improvement of the final published version. The protocol suggests counting electronic votes using their protocol to save time, then apply a Risk Limiting Audit (RLA) methodology when the post office mail votes arrive to check if there is a significant error. This limits the delay cases to those when RLA flags an error.

[7] Credit of the Wisconsin example belongs to Chain-link educational hub (https://chain.link/education-hub/blockchain-voting-for-elections), which reminds of the fact that the general idea of this solution is applicable on any trusted public permissionless chain (like Ethereum in Vocdoni design in [21]) not just Bitcoin.

that saw the actual true sheets. Even if the EA tried to change the finger print before being posted on the blockchain[8], a photocopy or mobile copy with their digital signature (using the EA secret key) cannot be claimed fake with overwhelming probability.

J. Mexico

As will be discussed in section 5.2 (V.B), Voatz successfully conducted online voting for nearly 234,117 citizens[9] in the Mexican 2024 election; this special case election had contests for President and Senate, and some states had contests for Governor and/or another local office, all in one ballot.

III. Voices Rejecting Dseploying Blockchains in Voting Systems

In this section, we will go through the main arguments of all researchers who wrote against the use of blockchain in e-voting through the years, [86, 91-93, 107-112], to discuss which of those arguments are still valid and in which cases. However, we will skip points against online voting in general since this is not the scope of the paper; there are many cases when blockchains are used with poll station voting as was discussed in section 2.

A. Controlling the Chain (Public Permissionless blockchains)

The main doubt raised by Josh Benaloh in [86] is that when it comes to major elections, we cannot just trust anonymous miners and/or full nodes to not add or alter transactions. This includes the commonly circulated risk of an adversary dedicating huge resources to attack the network. The possibility he mentioned of miners having some biases or conflict of interests towards a certain outcome (ex. voting on a candidate calling for a high tax on miners) can be generalized as the case of colluding/malicious miners or node operators in general; the cost and handling of the attack here will be different since those already have running nodes (no setup cost involved). In what follows we will examine the feasibility of each of these threat, but first we need to preface with some related information. Through the analysis, some issues raised by [91,93,101,107,111] will come up and hence will be also discussed.

1) Preface

Let us set the ground with some related definitions, numbers and rationales.

Computational Security:

When analyzing systems security, the term "*Computationally Secure*" considers the system secure if attacking the system is computationally infeasible with overwhelming probability; i.e., it takes a lot of money, time, and/or computation power to penetrate the system than an adversary could possibly have or find it worth the gain from the attack. Specifically, in nation level e-voting systems, the amount of resources (money and computing power) an adversary may dedicate to an attack depends on whether it is just a party competing to win, or a powerful country that has farsighted interests in the elections result; naturally depends too on the country and election type. The adversary's budget will differ whether it is a local parliament election, a presidential country election and which country, or a European parliament election in case of European countries; also, whether it is a peace or war time (the most expensive cyberattack maybe cheaper than war physical weapons)[10]. Having that said, let us look at the numbers.

Numbers:

-In USA, where most of the rejecting voices came from, the average presidential campaign funding for one candidate [111/ref.48, 123] was roughly $768 million in 2016, jumped in 2024 to ~ $2 billion by candidate and a total of $8 billion raised by the campaign. Agreeing with [111], this means that the adversary budget to disrupt or change the result of such election should be estimated to reach that number.

-The cost of possessing at least 51% of Bitcoin mining power was estimated by a Harvey Business School study in 2025 to be at least $6.07 billion ($4.6b for devices + $1.47b for electricity and maintenance), assuming the availability of the needed electricity and mining devices [124]; newer 2026 rough estimates raise this number between $8 billion [125] and $10 billion [126].

-The corresponding threshold for most Proof of Stake (POS) chains, including Ethereum since Sep 2022[11], depends on the total staked currency; as a theoretical lower bound, it needs more than 1/3 of staked amount. As of Sep 2026, [127], total staked Ethereum is 42.632959 million ETH ~ $105 billion which means an adversary must first stake more than $35 billion worth of ETH to think of controlling the network. Here, the cost of running the needed nodes (validators and block builder/proposer nodes) is far less than Bitcoin and can be ignored compared to the staked value.

-Another interesting numbers are those of power distribution between pools as captured from live monitors (Sep 2026). In Bitcoin, [128], the 10 days average shows Foundry USA pool mined 26.087% (390 blocks) followed by AntPool with 17.726% (265 blocks); note that these monitors are approximate or rough estimates, as they cannot capture existing hash power that failed the competition and did not mint a block. In Ethereum on the other hand, where people can stake smaller shares through large staking pools [129/pools/], [127], the largest pool Lido holds 21.49% (decreased from 27.94% on a prior check at 14/1/2025) followed by Binance with 7.6%.

-As for geographical distribution, live monitors on Bitcoin [130] shows %31.6 active full nodes in USA followed by 14.85% in Germany (~ 40% in Europe) in Sep 2026; while a validator deanonymization paper in 2025, [131/Fig.9] discovered 46.79% peers in Europe and 33.01% in USA.

Decentralization:

Although decentralization is an essential main feature in blockchain design, the ability to participate and the degree of participation still depends on the ability to afford resources. Hence, researchers have long studied the degree of decentralization for each chain through different performance metrics; [132] is a Systemization of Knowledge (SoK) 2025

[8] We mean replaced the TX with another one with a higher fee to get minted first (RBF in Bitcoin when OP_Return method is used like here or simply putting a much higher fee if any other blockchain is used to make the original one look like a double spending attempt). In any case, the attendants still have a copy with the certified digital signature.

[9] As an indication to the system scalability, Voatz says they though had to upload the records of ~100 million eligible voters into their servers.

[10] Recall in section II.B (2.2) Russia suspending its e-voting system due to war [52].

[11] Before Sep 2022, Ethereum was Proof of W\ork (POW) chain like Bitcoin; the reader should put in mind this fact when considering any estimated numbers or written arguments before that date.

paper that summarizes previous research, then study different metrics on different blockchains and provides an interactive live dashboard for them [133]. Also, [131] captured another interesting angle when discovered that many validators in Ethereum (could similarly be traced for Bitcoin miners) distribute their power among a number of pools creating dependencies between pools and discusses the implications of that [131/sec.8.4] on the degree of decentralization; i.e., there is another dimension and the above ratios in live monitors are not enough alone.)

Censorship Resistance & The Anonymity Property

Anonymity in blockchains means that users generate a private-public key pair, what is called Public Key Infrastructure (PKI), and are identified only in the transaction through their public key address while they get to keep their private keys; in general users are advised to generate a new pair for each transaction. Hence, one should expect a first line defense for election authorities is to keep hiding from adversaries through a chain of addresses and keep generating new ones for every step. However, things are not that easy; even if election applications did more than and tried to use encrypted transactions and/or hide using things like Transaction mixers and Tor applications; challenges remain high, [134], and the IP address generating the transaction can be identified in most cases. Another approach is to use privacy focused blockchains like Monero and Zcash [135]. Still, real life examples of this approach can be found in crypto crime reports, [136], of criminals and sanctioned countries who tried to do that; yet, tracing remains possible especially with higher adversary powers.

2) Bitcoin

1-If an adversary invested all this money to perform a double spending attack or to reverse the last unconfirmed 6 blocks (cannot reverse or alter earlier than that), the money gains (in Bitcoin currency BTC) will be almost negligible to the cost incurred; this will also shake trust in the currency making the gained value decay further. The author in [124/ sec. 4.A] argues that the adversary could invest in perpetual/conventional futures that rise when BTC price goes down; the author then added that nefarious actors might target the network for other reasons[12], which exactly describes the profile of election adversaries.

2-A defending argument is that attacking Bitcoin is not just about money, the needed electricity is a *huge physical barrier*; even at the nation level, a country (say Russia/USA) gathering the needed electricity can hardly go unnoticed by major intelligence agencies of other countries (say USA/Russia respectively). Although [126] mentions the existence of rentable hash markets "*Nicehash*" which may facilitate a hit-and-run attack, it clarifies that the "*Nicehashable*" metric does not exceed 5% in Bitcoin; hence, it is not enough to break the physical barrier. However, **what if an adversary distributed the mining devices (and hence the needed electricity) into say 20 different nodes in 20 different countries, will gathering the electricity be as noticeable by intelligent agencies?** We believe the answer remains debatable.

3- For the possibility of biased miners or mining pools, things get more risky as the live ratios of active full nodes, [130], and pools [128] reveal; the threat increases with newer research findings. For example, the authors in [137] use a game theoretic model to argue that a double spending is possible with only 26% majority; this alone halves the adversary's budget, what if combined with bribing miners and rentable hash. Actually, it seems a feasible scenario to imagine a competitor or an adversary in a USA presidential election achieving the 26% barrier (or even 51%) by splitting his/her budget between buying people and buying devices with some renting if possible. Although not that feasible for small developing countries that may benefit from the idea of immutable democracy, opposing voices may doubt foreign interference from USA or Europe who control a significant ratio of mining pools and active full nodes. However, a defense maybe that those claims already exist with or without the blockchain.

4- As for USA and similar powerful countries, it is expected for every concerned entity to run a Bitcoin full node in that case leading to attack detection where the truth remains a consensus between those entities. Governments can go a step further and run a significant ratio of nodes during election time; this was described by Ben Adida in 2017, [107], in the case of handling just meta data (like fingerprints) as solving a small part of the problem with an *unnecessarily big hammer* and by Nir Kshetri in 2019, [93], for the general case as unnecessarily exhausting the government budget.

3) Ethereum and POS Blockchains

1-In addition to a starting budget of more than 1/3 of the total stake ($35 billion in case of Ethereum [127]), POS chains deploy extra techniques, [138], to limit the attack effect by detecting and slashing the stakes of malicious nodes in the next epoch. Meaning the adversary here will lose a significant portion of his/her stake if the attack failed; this raises the threshold to 5/9, [139], of total stake for the attack to keep going. Meaning that the loss after a failed attack will be more significant if compared to Bitcoin case (POW in general) where a failed adversary will lose only the electricity and maintenance cost (~ $2 billion) but still have the mining devices (~$5-6 billion) to keep or sell.

2-In addition, recent studies, [140,141], shows that higher lower bounds can be achieved in reality when considering which staking nodes (accounts/addresses) are currently active and participating in the block proposing-building-validation process; this can dramatically raise the lower bound to 99% of active stake in some cases. Considering the slashing mentioned in 1, the attack resistance may increase further by calling honest "*sleepy*" clients as [141] describes them to actively participate after discovering a malicious address and slashing a portion of its stake. Comparing to Bitcoin (POW), governmental participation here will be through staking while the cost of running validator nodes here is minor; i.e., the invested budget will remain (only subject to ETH price changes) and may even result in some added profit.

3-As for existing validators/pools who may have a conflict of interest (certain bias towards the election results), or the possibility of bribing them, we consider two threats. If they tried to jeopardize the liveness of the network (does not post a block when chosen as a leader) or to validate contradicting blocks/transactions, they will be slashed and lose a significant portion of their stake; also, traced as criminals and may become a subject to trial if in the same country. However, they

[12] Note also that, unlike other blockchains that allow smart contracts with programming capabilities, Bitcoin allows only a limited number of scripts (TX format). Consequently, this limited its usage to mostly money transfer TXs; i.e., haven't faced many application cases where hackers' targeted gains are not just than financial.

can still post other blocks that do not contain election transactions (if delayed long enough may raise some doubts by losers), or even worse manipulate the transaction without being discovered if kept enough majority for a few epochs. We have already mentioned in [131, 132] that concerns are already discussed on the literature regarding the degree of decentralization and concentration of power; [142] is a newer 2026 study on the risks of staking pools when there are malicious agents.

4-On the positive side, most POS chains invest a lot of effort and money in software auditing and in making the leader choice of a certain block purely random [143]; meaning that the adversary's chance in controlling a certain block at a specific point of time (an election fingerprint posting for example) remains probabilistic and pretty much unlikely for a same cluster to control consecutive epochs. In addition, Ethereum is a continuously evolving network that keep fixing exploits; for example, the authors of [131] declares that they received a bug bounty for their findings and the ratio of the leading pool "Lido" decreased ~ 6.5% last year.

4) Controlling Private Permissioned Chains

Although this section is mainly about the threat of controlling major large-scale public blockchains like Bitcoin & Ethereum, we point out that the monopoly risk is naturally more feasible in private permissioned chains. When a custom private blockchain is designed to take the place of the traditional public bulletin board (PBB), it usually consists of a limited number of servers that validate transactions according to some consensus rule. Features examples include zkVoting (South Korea) [75] that runs Ethereum protocols on a local Ganache environment and hence consensus (the cryptographic rules on which transactions are validated) is done according to a quorum of trustees. Also, Voatz (USA, Canada, Mexico) [96/sec.5, 97/Blockchain servers] that uses 32 Linux servers split equally between Amazon AWS and Microsoft's Azure; hence, it remains vulnerable to possible control by Microsoft or Amazon if not by surreptitiously adding resources as mentioned in [96,97] could be through trapdoors or exploits in the servers or their operating system[13].

5) Concluding Remarks

In this subsection we have gone through the two major chains Bitcoin & Ethereum; also cited a proposed research article that uses a privacy focused chain. The reader should be aware now that monopoly threats do exist, but still have a defense and/or protection strategies that makes them debatable most of the time. Countries and involved entities are expected, and advised, to invest some money (stake/hash rate) in the used blockchain to gain a safety margin power threshold. Finally, it remains a different decision for each country to estimate the power of its adversaries and weighs the pros & cons of each choice. The scientists in USA for example fears money power intervention the most because estimated adversary's power is almost the highest worldwide, and see paper ballots as a safe option, [111], because they are not faced with problems like those reported in the video about Guatemala [85]; i.e., there is no single recipe that fits all countries.

B. Software Dependency (Attacks &Vulnerabilities)

We start with some software related terminology definitions, two of which are included by the MIT paper [111] in the 5 minimum necessary but not sufficient requirements for secure elections in an evidence-based framework; ballot secrecy, software independence, voter-verifiable ballots, contestability and auditing. Then, we move on to discuss the software related concerns (code vulnerabilities) and the defending arguments against them before deducing any conclusion/recommendation at the end.

1) Terminology & Definitions

Software Independence [111, 111/ref.35,36] is defined as *the property that an undetected change or error in a system's software cannot cause an undetectable change in the election outcome*.

The definition clearly contains software attacks & vulnerabilities and extends to include any deliberate code alteration say of an opensource election code for example. Whenever there is a dispute about the existence of an error/alteration, another metric is defined for elections.

Contestability [111,111/ref.37] *A contestable voting system is one that provides publicly verifiable evidence that the election outcome is untrustworthy, whenever an error is detected.*

This property relates to providing a publicly verifiable audit trail to the election results; a cryptographic proof (usually a zero knowledge one to keep vote secrecy) that the submitted list of ballots is the same counted list of ballots.

Transparency [111] Transparency in voting systems usually refers to auditing the election process in all its steps by candidates' representatives and neutral observers (civil society, local and international NGOs, ...). However, in electronic voting systems, transparency also includes *disclosing the system design, the cryptographic protocols used and open sourcing the election code (with documentation)*.

-Many existing systems lacked such property that researchers had to apply reverse engineering on them to get some clues; examples include the Russian voting system [47,144] and Voatz in USA [92,96-98]. In fact, we believe most blockchain rejecting voices in USA were originally motivated by their suspicion due to the secrecy of Voatz design; still after reverse engineering, [96], the authors could not determine what is exactly stored in the blockchain and had to optimistically assume that all cryptographic activities are logged to the blockchain, and that these blockchain records are secure, monitored, and immutable. Recall that the title of [96] clearly states that "*The Ballot is Busted Before the Blockchain*" and the abstract describes their work as a "*concrete illustration of the common wisdom against internet voting and* ***the importance of transparency***".

-Also, a related issue here is to prove that the open-sourced code (if existed, hence analyzed and approved by involved entities) is the one actually used during the election (i.e. has not been altered).

2) Concerns

[13] This is an equivalent threat to buying Ballot Marking Devices (BMDs) manufactured in another country; watch this 2017 lecture by Dan Wallach (https://youtu.be/CI8q_356zZs) where he discusses possible threats even if the BMD is isolated.

1-The scientists and associations in 2018 [91,101] were concerned about the fact that blockchains like any software can have exploits and live a continuous evolving battle; from cryptographic attacks to design to simple software exploits when thousands of code lines are written. When people say we use it every day in our financial transactions [], the MIT paper in 2021 [111] replies that attacks do happen in financial cryptocurrency transactions and even in digital banking systems, so why risk it in elections when the adversary' power could be much larger and the chance of tracing the thief or retrieving what was stolen (your vote) is far less in elections. In this context, we bring up a famous 2016 incident of *Ethereum reorg*, [145/DAO fork,] to return more than 3.6 million ETH that was stolen in a hack[14] by withdrawing the money from the original smart contract (transaction) before the hack time; most election fraud scenarios will not probably need a *reorg*, but it might be harder to reach consensus on which transactions are fraud and which are the correct ones.
2-The paper in [111], and the US Vote foundation in [112], point out to another risk about **the attack scale**. While an alteration of paper ballot boxes is physically noticeable and limited in size, an alteration of a few lines of code could possibly change millions of ballots and is harder to detect (recall the related issue to transparency).
3-The authors argue that software independence should be **checkable using non-software means**; they compare to paper ballots being physically available for recount (comparing the strength of a hardcopy vs. softcopy document as an evidence/proof).
4-An attacker willing to spend enough money could flood the blockchain with transactions, [111,112] to drive up fees and keep users from voting until after the deadline passes.

3) Defending Arguments

1-Of course, Ethereum, and most smart contract enabled blockchains that have an Ethereum Virtual Machine EVM design, are more susceptible to newly discovered attacks or software vulnerabilities when compared to Bitcoin primitive script language. For that reason, they endure continuous updates for improvements and/or fixes (hard forks [145]) unlike Bitcoin that rarely invoke such option.
2-In all significant blockchains, adding or updating any piece of code goes through many software auditing and testing phases before the actual integration; besides, it is all open-sourced. As stated earlier, many USA rejecting voices were partially influenced by the lack of transparency in the used system (Voatz); this does not have to be the case in other systems. In fact, open sourcing the code remains a debatable issue even in systems that do not use blockchains; for example, the voting application in the Estonian voting system [146/Assessed Threats/(Open Code, Closed Code)].
3-Cryptocurrency defenders think of major cryptocurrency blockchains like Bitcoin and Ethereum as having the "*Too Big to Fall*" property. They hold a lot of applications, a lot of money transactions and a large market cap; attacks/exploits can be rapidly discovered and traced, [136], even if sometimes it becomes hard to return the stolen money. Election-wise, when there is no money to retrieve, an attack or any kind of manipulation or network monopoly can hardly go unnoticed; i.e., like any other voting mechanism, a worst-case scenario is a recount or re-election. In fact, especially in hybrid systems when used with poll station voting as in Georgia [119], they may better help in locating the error location (identifying the faulty partial sums) to minimize the recount efforts (the examples from previous elections in [121,122]).
4- Specifically for election auditability and contestability, all electronic voting systems, including those who deploy blockchains, include a publicly **verifiable audit trail** using NIZKPs (Non-Interactive Zero Knowledge Proofs). Most of them also apply some kind of physical checking of a random sample of votes using statistical error measuring techniques; i.e., **Risk Limiting Audits (RLAs)** [115/audits/whatisrla/, 147].
5- An important argument that we partially disagree with [111], is that paper ballots are not necessarily more **voter verifiable** and/or coercion resistant; yes, a hand marked paper ballot is unbeatable evidence on "*recorded as intended*" but scores much less than electronic voting in "*counted as recorded*". There are many reported cases in many countries (ex. Guatemala in [85]) where complete ballot boxes could be altered. Regarding **coercion resistance** and **ballot privacy**, malicious poll workers or crowded poll stations may jeopardize vote privacy and other aspects as well. In addition, if coercers could get just one empty paper ballot, there is a technique called "*circulating ballot*" where coercers/vote buyers hand the voter and already marked ballot as they desire to throw in the ballot box and require him/her to give them the new empty paper ballot to repeat; the best the coercer can do here is destroy the marked ballot and not vote at all. We agree that both internet voting and blockchains add new attack vectors (especially in vote buying/coercion as will be discussed in section), but we also point out that pen marked paper ballots have their major issues too.
6-We believe e-voting systems featuring those posting cryptographic fingerprints on the blockchain like [20-22,83,85] scores much higher on **contestability**. No voter could ever verify that his/her pen-marked paper ballot is included in the final count; how would a losing candidate (or election observer) prove that the partial counts he/she witnessed led to different values than the published ones when draft paper photocopies are not enough as a proof in court. For those cases, regular electronic voting systems provide QR code proofs which may not be enough evidence in case of dispute, while blockchain-based ones provide immutable publicly available proofs; you may doubt those software cryptographic proofs and/or worry about the hidden exploits, but you do not have an equivalent option.

4) Concluding Remarks

We believe the concerns raised by academia and associations do not include the case of just adding an extra security layer to the existing systems through posting cryptographic fingerprints of partial results on the chain [85,119,120]; the authors of [111] state clearly that they are focused on internet voting and that using blockchains to help in other aspects of election like auditing is not covered in their

[14] There was a voting on that with 85% approving (https://blog.ethereum.org/2016/07/20/hard-fork-completed); those who objected insisting on keeping the principle no matter what (Code is the Law) forked into another cryptocurrency (Ethereum Classic).

paper. Then, we again argue that it is a case-by-case basis decision for each country to weigh the pros & cons. In general, we recommend posting partial timestamped fingerprints (the *SimpleProof* approach [120]) for countries that use poll station voting. Especially those who use pen marked paper ballots (no voting devices), this can provide a significant security improvement with minimal cost[15]; careful timing should be chosen to prevent prior alteration, and the software calculating the cryptography should be opensource and audited by involved entities. If done right, with enough auditing to make sure no manipulation was done earlier, this can make any voting system scores better on verifiability and contestability. Also, we recommend complete transparency in e-voting systems design; the architecture, the protocols and the code.

C. Coercion & Vote Buying

We have elaborated in the previous subsection that pen-market paper voting systems have their own coercion and/or vote buying risks that are not possible in online voting; even if the coercer is living with the voter as [111] says, the odds of sneaking a moment away or with different device remain higher than in the fixed short time in poll stations. However, we did not deny though that a new set of vulnerabilities come with online voting and another may result in case of using blockchains. This section summarizes those threats raised by [92,111,112].

1) Threats that are general to most verifiable e-voting systems

1-A straight forward threat for all internet voting systems is compromised devices or compromised communication channels; [111/systems attacks] discusses that threat in detail with real world examples. Existing i-voting systems like those in Estonia, Canada, France, and Switzerland have it in their regulations that a compromised voter device must not be enough to forge a vote; however, we are here concerned about revealing the true vote to the vote buyer/coercer.

2-In addition, whenever a receipt or a QR code is provided for voters to verify their votes, it can be used by vote buyers and/or coercers as well. Many e-voting systems mitigated this dilemma by allowing multiple voting [148]; you can simply show the coercer the old receipt/QR code then vote again as you wish. This can be defeated if the coercer can find out somehow all your voting transactions or the last one (or its timing) or can force you at the last final moments before voting is closed.

3-Section 5 will discuss new vote buying/coercion threats that resulted from the existence of blockchains, even for e-voting systems that do not use blockchains.

2) Threats emerging from blockchains deployment

-In 2018 Ari Juels and his colleagues, [92], expressed concerns that blockchains make it easier for vote buyers; an experience they realized from cryptocurrency related small scale e-voting in Decentralized Anonymous Organizations (DAOs) through governance tokens.

-On that time, they were working on an idea about vote buying threats they called DARK DAOs [149]; they showed how conditional payment to the vote seller (automated via a smart contract to his/her wallet) can be triggered by a remote execution attest from a Trusted Execution Environment (TEE) that a certain code has been executed[16]. So, if a certain e-voting system assigns voting rights to voters as tokens or coins, [111/coins as votes], the existence of Trusted Execution Environments TEEs, 2019 [150], on the voter device may facilitate buying or tracing that token[17].

-The research group pursued the thread further in the recent years, 2025 [151-153], to demonstrate how a token or its access rights can be rented secretly for a fixed interval (the election days) which facilitates vote buying. Of course, this threat does not exist with poll station voting whether using pen-marked ballots or using voting machines when posting cryptographic fingerprints on a blockchain (no tokens or TEEs are involved).

D. Malleability Threat

1) Preface

-Malleability as a property is mainly related to the chain deployed cryptography; if two different plaintexts can result in the same cryptographic value (hash/digital signature/encryption/….), or can have more than valid cryptographic value, then an adversary can replace one with another. Many e-voting papers, [39,154], either prove or start by assuming that all cryptographic protocols used are Non-Malleable against Chosen Plaintext Attacks (NM-CPA); the most common approach (goes without saying when using shuffling mix-nets for example) is to use NIZKPs [155]. Blockchains also have their malleability problems that may affect the e-voting systems using them.

Transaction malleability:

-Blockchains, and many e-voting systems, deploy Elliptic Curve Cryptography, where cryptographic functions can be equivalent to calculating a point on the elliptic curve used.

-It was first noticed by **Bitcoin**[18] , [156], that the mirror point on the other side of the axis could be considered also as a valid signature. Putting it in the simplest possible way, since calculations are all done modulus a very large number say N, any given signature can be represented as (R, S) or (R, N-S);

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[15] even in Germany as a developed country, the cost reduction achieved by *opn.vote* pilot [22,41] was highlighted as one of the main advantages resulting from using Ethereum-based online voting (reducing the costs per vote from €2 to under €0.01).

[16] Remote execution attests were originally discussed on Intel SGX (which is available on many new PCs in the market: https://www.intel.com/content/www/us/en/architecture-and-technology/software-guard-extensions-processors.html), and it exists in other processor companies like Apple and others as well. So, we can assume there's a considerable probability that the voter PC supports it.

[17] Vitalik Buterin et al. introduced a concept they call "**CK" *Complete Knowledge*** that tries to make the trusted hardware sends attestations only to unmalicious safe code (authenticates itself); *Complete Knowledge: Preventing Encumbrance of Cryptographic Secrets";* https://eprint.iacr.org/2023/044. However, this won't help if the voter device is totally compromised (the execution attest is used to protect the coercer from the middleman/dark web cheating on him), or if the voter is voluntarily selling his/her vote.

[18] A subset of the general malleability problem was first solved in Feb 2013 released update, Bitcoin core 0.8.0, by accepting only new TXs with digital signatures following Distinguished Encoding Rules (DRE) format, Feb 2015 core 0.10.0 made it strict (https://gnusha.org/pi/bitcoindev/55B79146.70309@gmail.com/); the point from the previous link about OpenSSL is for the reader to understand how fixing and/or discovering malleability cases can get quite technical and is specific to each chain. The interested reader can

this makes it possible to change the transaction ID of a confirmed transaction which may be used to steal coins. Finally, in 2017 the *Segregated Witness (SegWit)* prevented Bitcoin transaction malleability, among other enhancements like scalability regarding the block size, when it separated all transactions signatures (usually called cryptographic witnesses) at the end of the Block.
-As explained, the solution is not general to all blockchains; for example, the **XRP** ledger (**Ripple**) [157] completely solved malleability for single-signed transactions in 2020, while threshold multi-signature transactions can still have malleability issues if they have more signatures than the necessary threshold (M-to-N threshold homomorphic signature).

Merkle Tree Malleability:
-In addition to digital signatures, Merkle Trees are another widely used cryptographic construction, whether by blockchains or by e-voting systems even without blockchains. A merkle tree is built from all hashes (transactions in a blockchains' block or could be ballots in an e-voting system); the tree can be used to derive a cryptographic witness on the existence of a certain leaf node where the root value is included in the block header of most blockchains. We can describe merkle tree malleability as the ability of an adversary to alter the tree nodes in anyway while the Merkle root remains the same (ex. replace an intermediate node with a fraud leaf) without being detected; the authors in [158] (2024) discuss the security implications on blockchains in general.
-In **Bitcoin**, there was an old merkle tree malleability, CVE-2012-2459; if the number of transactions is odd, the last value is duplicated which means that repeating the last transaction twice will lead to the same merkle root value. This was fixed in Bitcoin core 0.6.1 2012, [159], by rejecting a transaction with a hash value already existing from inclusion in the block. Another one that could lead to deceiving users with fake inclusion proofs for their transactions was fixed recently by BIP 54 (https://bip54.org/).
-Another case example is a famous attack, [160], on a Merkle tree library widely used by developers in Ethereum, *OpenZeppelin*, that targeted the way Merkle root is calculated. Note that the vulnerability is quite similar to what was fixed in Bitcoin 2012, yet it was invoked again in an open-source library due to the code implementation method.

Concluding Remarks:
Most large-scale blockchains have mitigated their transaction malleability problems in a way specific to their transaction format and/or design. Hence, the threat remains significant only when customized blockchains are designed for the e-voting system without handling/checking the existence of such vulnerability or using the famous ones in a way that could revive the exploit. The threat exists also, for e-voting systems that use these cryptographic constructs even without blockchains either by mutating an individual ballot or a group of ballots; for example, a hypothetical e-voting system that depends on a Merkle tree of the ballots lists for proving votes integrity (individual and universal verifiability) should check malleability risks of the used Merkle tree implementation.

2) Transaction Malleability Threat to e-Voting systems
The general threat, as mentioned above, has many possible scenarios. These two papers, 2019/2021 [161,162], conducted simulation and testbed experiments concerning transaction malleability in e-voting applications. The authors created a mutated malleable transaction T_m that alters certain fields in an honest transaction T_h and assumed a Bitcoin-like POW environment. To simulate the adversary's hash power, they made building a block containing T_m easier (solving the nonce puzzle); they also injected a delay function for the honest transaction T_h to simulate any random network latency or malicious deliberate postponing (recall denial of service concerns discussed in the previous sections). In other words, the attack success would be a consequence of the adversary's success in the threats/concerns discussed in the previous subsections (A&B); deliberately delaying the honest transaction and/or controlling the network to put his/her block first.

IV. Blockchain Facilitated Attacks on All Voting Systems

We discussed in section 3.C how deploying blockchains can make coercion and/or vote buying easier. In this section, we show how the new technology provided coercers and vote buyers with new tools and facilities that they can use with all kinds of voting systems.

A. Smart Contracts and Automated Online Code Execution

1-We previously discussed how TEE execution attests, or verifying QR codes entered through an interface, can be used to trigger a smart contract, [149-151], for **automated conditional payment to the vote seller's wallet**. This can happen equally likely for any online voting system even if does not deploy blockchains. The smart contract here provides a guarantee for sellers that they will get paid, and the anonymity property makes it easier for the buyer/coercer and sellers to hide from the authorities.
2-Blockchain online auctions, where buyers place their bids (anonymously) online, happen all the time; there is no infeasibility condition that prevents automated **online vote auctions**, [163], from happening between anonymous competitors (candidates and/or external adversaries) during election time. Different scenarios include vote sellers directly connecting to an interface site that will execute the conditional payment described above afterwards, Fig.4, or a dark web intermediary who controls many compromised vote credentials and will be paid accordingly; this may also happen (but less effective) in poll station voting to provide better timing. One could imagine, in all scenarios, the auction running till the last minutes (like race/match results bidding) where bids and votes' prices get higher with time.

find the exact sequence of events and links to discussion traces in (https://en.bitcoin.it/wiki/Transaction_malleability#Signature_Malleability)

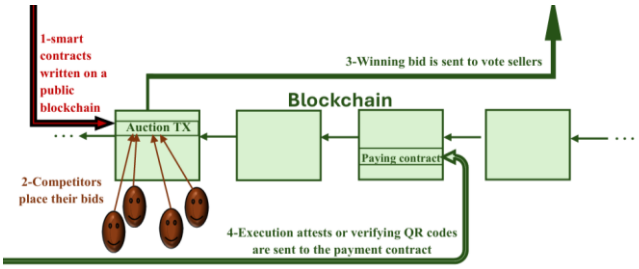


Fig. 4. A possible scenario for online vote auctions using blockchains.

B. *The Immutable Public Ledger Property*

The authors in [164] show how new technologies, including the presence of a public blockchain immutable ledger, can help adversaries (coercers and vote buyers) in forcing voters to comply or help voters to obtain a (probabilistic) receipt for vote selling. The paper examines the vulnerability of the important known e-voting systems with a main approach that resembles Benaloh challenge in asking the voter to act probabilistically but here according to cryptographic randomness either produced by the voting protocol or by the blockchain. For example, post the vote or its hash/cipher on the chain, then spoil the vote if the next block number starts with zero and show me the receipt; if the voter lied and posted a wrong value at the chain, he/she might get caught with probability 50%. The paper, however, assumes the voter can post a long cryptographic bitstring value to the blockchain while in booth; i.e., the threat is more significant for online voting where the voter can copy-paste those values and have enough time to post on the chain.

V. Working Case Examples (South Korea, Voatz)

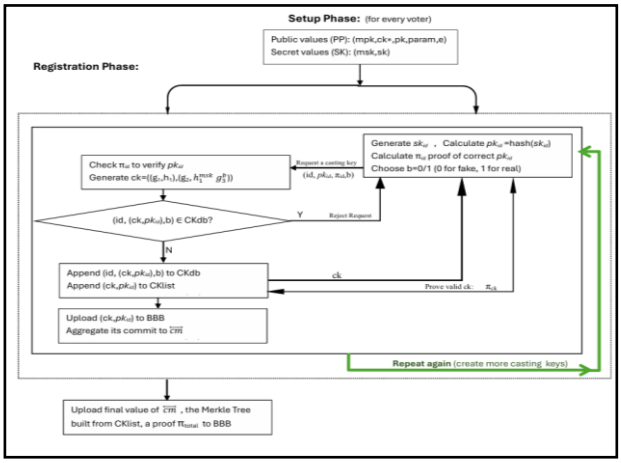
In this section, we will go through the details of some case examples and examine them against the concerns in the previous section.

A. *South Korea zkVoting*

A joint work between *Zkrypto* company, Hanyang University and Kookmin University, [165, 166], won nation competitions for two consecutive years, (CES2023, etc.) [75,167], reaching this final used design.

1) *Main Design*

While the flowchart in Fig.5 shows the detailed steps, this section grasps the main design ideas.



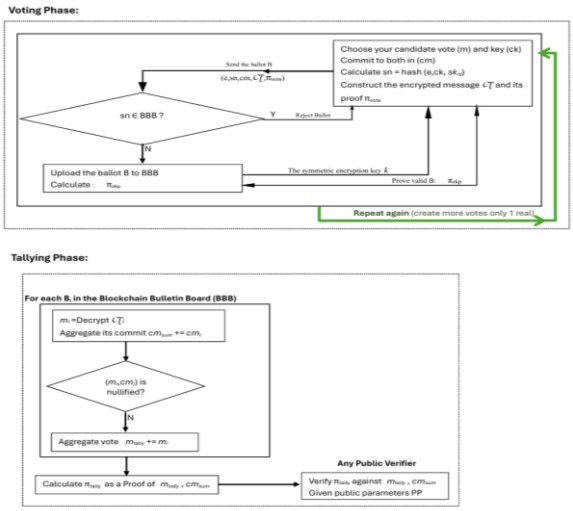


Fig. 5. The main steps of the zkVoting system, some cryptographic details regarding random numbers and relations were skipped for simplicity without affecting the core protocol design. A clarifying note is that ck can have exactly 1 value when b=1 because g_3^b will equal g_3, while g_3^0 can have different values in this context.

-Voters are provided with one real casting key (ck,1) to cast their real intended vote and as many **fake casting keys** (ck,0) to deceive coercers with fake votes; all votes (fake and real) get committed as transactions on the blockchain (assumed as the PBB, sometimes referred to as BBB in their paper adding an extra "B" for Blockchain) where only the tallier can distinguish the real vote from fake votes. The voter has another **nullifiable commitment key ck*** which is used to nullify keys as fake and encrypted ballots of fake keys open to zero value when aggregating all votes.

-Voters are provided with **digital receipts** linked to their voting transactions on the blockchain using a serial number (calculated by the voter from his/her keys); the receipt includes a symmetric encryption key to decrypt the ballot (like the QR-code could be verified on any device).

-Another cryptographic proof is added to **verify the total number** of real keys equal the number of aggregated commitments after nullifying fake votes. All keys (fake & real) are stored in a list where *proof of set membership* is applied before approving the ballot; in their implementation [165] Merkle Trees are the used construct.

2) *Fake Votes (Fake Credentials vs. Fake Casting Keys)*

The idea of deceiving coercers with **fake credentials** was originally introduced in 2005 in *JCJ* [165/ref.11] and a modification of it was implemented in *Civitas* 2008 [165/ref.12]. The introduced system, *zkVoting* [165], used fake keys instead of fake credentials for two purposes:

1. Removing trust from the registrar since the entity generating the keys is different from the entity collecting the votes; i.e., attacks can only happen if the registration application and the voting application colluded together.
2. Decreasing the time complexity of the tallying phase (distinguishing the real vote from $O(n^2)$ to $O(n)$ where n is the total number of ballots fake and real) since the use of nullifiers with fake keys will make fake votes open to zero value.

3) *Nullifiers and Nullifiable Commitments*

Nullifiers were first introduced in *zerocash* protocol 2014, [170], which aimed to improve on zerocoin[19] (shares 3 authors with the one behind Zcoin [170/ref. MGGR13]) and eventually evolved to the privacy focused cryptocurrency ***Zcash***. What the authors called *serial numbers* at that time and used to mark the spending of a coin (i.e. to prevent double spending afterwards) was globally called *nullifiers* in further research[20]; the main design architecture of a nullifier set and a commitment tree became a standard approach in many privacy preserving chains. Looking into Fig.5, *zkVoting* also uses CKdb list of all (ck,pk_{id}) and a Merkle Tree to provide proof of set membership[21]. Other uses of nullifiers emerged with time like validator "*unlinkability*" in Ethereum [171].

Regarding e-voting, the main concern of this paper, we trace the first usage of nullifiers for coercion resistance in a master thesis (2022) [172] and a follow up paper (2025) [173]. The equation (15) in [172/sec.4.1.2] is quite similar to how ck is calculated in Fig.5 (the registration phase); however, there are a few significant differences:

1. nullification here happens after casting the vote only in case of coercion. In other words, the main semantic difference and/or novelty in [165] is that fake keys are determined from the beginning at the registration phase, while [172] follows the same theme of [170] that nullifiers may or may not be added according to the user spending pattern (voter coercion circumstances). As a consequence, the total number of nullifiers in [172,173] could be used by election observers as an estimate on general coercion/vote buying attempts, while the earlier generation of fake keys in [165] is not; the **coercion evidence** idea elaborated in [173] is originally due to *Caveat Coercitor* [173/ref.18] which is an 2013 advancement on JCJ (2005) and Civitas (2008). In fact, the coercion resistance strategy of *zkVoting* resembles *Decoy ballots* [173/ref.7 & sec.1.2 &Table1] in the sense that they are requested during registration (prior to voting), although differ in using a cryptographic approach nullifiable commitments instead of a game theoretic one.

2. Also, the protocols in [172,173] do not use **blockchains** and are introduced as an overlay, add-on, or "back-end" for the "front-end" of any E2E coercion resistant voting scheme; this does not prevent adding them with blockchain protocols though. In the thesis, [172] was implemented over *VoteXX* system [172/ref.17] improving on Mix and Match protocol [172/ref.39] concentrating on computing binary XORs under encryption; the overlay of [173] was illustrated on VoteXX and on JCJ [165/ref.11] to only need an additional O(n) at tallying time with some cryptographic tricks and concurrent processing.

3. The main strategy of [173] is to allow each voter to declare a trusted helper (named *hedgehog* strategy) who can nullify or flip votes if the voter was under great pressure by coercers and cannot do it by him/herself. It is our belief, the authors of this paper, that the whole idea collapses when considering the possibility of **a coercer/vote buyer forcing voters to be their trusted helper**; pretty expected from large-scale vote buyers in authoritarian regimes. This cannot happen in zkVoting unless the coercer attended or controlled the whole steps of the registration phase.

4. The authors in [173] put **key deniability** as a future work, while *zkVoting* proves it is satisfied in [165/Appendix B/Theorem 2.1]; put it simply, unless a coercer attended the complete registration phase to know the value of bit "*b*" sent with each request of casting key, the coercer cannot know whether a certain key "*ck*" is fake or real.

Without getting into the details of those protocols, we clarify that nullifiable commitments, or nullifiers in general, is a zero knowledge proof idea that is not necessarily bounded to blockchains although first appeared there. In fact they are one of many examples where the fast evolving rhythm of blockchain technology, including zero knowledge research, enriched e-voting research with new tools and/or ideas.

4) Security Analysis Against Mentioned Threats

1-As for malleability, the authors assumed from the beginning that their nullifiable protocol is binding, [165/sec.2.1/def.2.3]; i.e., no two ballots can be committed to the same value. Formally, for a key k (real or fake) probability of [Commit(k,m_0;r_0) = Commit(k,m_1;r_1)] is negligible.

2-The execution attest and/or liquefaction of voting token attacks are not applicable here; the voter can deceive the coercer to receive the attest and pay the money while previously nullifying the used casting key from another device. Hence, a rational coercer/vote buyer will realize from the beginning this is not enough as a proof of compliance.

3-Both JCJ and Civitas were found vulnerable by the authors of [164/sec. 3.6] if the coercer forced the voter to begin with a certain public key K_{VE} before the registration starts. The authors here assumes a nullifiable commitment scheme should be hiding; in [165/Appendix A.1/Fig.3], even an adversary knowing the nullifying key ck^* and the master public-private casting keys pair, (mpk,msk,ck^*), cannot guess with probability more than ½ which keys the voter has nullified (whether a certain casting key ck is accompanied with a bit b=0 or b=1).

-Moreover, the protocol is assumed to have *indistinguishability of keys* to the commitment framework; this means that knowing two casting keys where (ck,1) is real and (ck',0) is fake, the coercer cannot identify from the commitments published on the blockchain (PBB) which key is used for each ballot (cannot guess with probability more than ½).

[19] The same *Zcoin* used in the Thailand pilot 2018; although Zcash existed at that time, the secrecy of the TX destination was not a problem for an e-voting application as it is always the Election Authority (EA), while the amount (could reflect the chosen vote) may have been hidden through randomization.

[20] footnote 2 in the latest Zcash protocol specification on 15 Sep 2026 (Versionv2026.7.0-191-g0fae78[NU6.2]).

[21] In [170], the user may spend a coin (corresponding to posting a vote using ck) by sending the nullifier and a proof π that "*I know r such that Commit$_r$(nullifier) appears as a leaf in a CRH-based Merkle tree whose root is rt*".

-However, similar to its predecessors [165/re. 11&12], **it all depends on a safe registration phase; if attended the registration phase, the adversary can know the real casting key (ck,1) and also the secret key sk_{id} used in the voting phase to cast the ballots.** If the coercer used those values to cast a desired vote (could be from another device) and got rejected, he/she will know the voter deceived them and voted with the real key to another choice. Related to this is what the authors stated in both presentations [75/video,165/video] that the government requested that the Election Authority (EA) generates all keys in advance (setup phase in Fig.5) to increase usability; we find real concerns regarding **Ballot Stuffing** in this case since registering in place of absent voters (and hence voting on their behalf) becomes possible. The authors talked about protecting ballots secrecy through splitting and randomization but did not discuss the threat on integrity; i.e., considering that voters can avoid such threat by participating in election, or considering EA to constitute a quorum of trustees [174/Quorum for decision] under threshold cryptography (cannot collude beyond the threshold). In any case, if the original voter id were included in the public Merkle tree CKlist posted on the blockchain, to be (id, (ck,pk_{id})) absent voters could be encouraged to check that their ids are not included (non-inclusion proofs are supported by Merkle trees); election observers and public verifiers could also check a random sample. In this case the *immutability* of the public ledger serves as an added value to secure the consistency between CKlist and posted Ballots; however, there is no way of telling (probability = ½) who is lying, the government or the voters.

4-Although related to online voting in general, not specific to blockchain deployment (could happen with any PBB), a fake voting application accompanied by a fake registrar can deceive the voter with a nullified digital receipt. Assuming the adversary A is an eligible voter, **with a fake application A can take the voter choice, cast a corresponding vote using his/her own fake keys and sends the voter the digital receipt**; this will kind of switch the places between the voter and the adversary making the system treat the voter as a possible adversary who has no right to distinguish whether the vote is real or fake. This can work only if the registrar was also a malicious rouge application (or if there is only one application for both phases in the real implementation as the demo in [165/video]) to send the victimized voter fraud credentials, some of which belong to the coercer, so that the voter finds sn=hash(e,ck,sk_{id}) and other cryptographic verifying values seems to be correct. This issue was questioned by one of the audiences in [165/video] referring to Ron Rivest previously mentioned concerns on online voting; the answer of checking the "sn" value is not quite sufficient here since the fake application can copy to the voter exactly all the adversary's correct keys except which key is fake. As for comparing with [173], we believe this attack is worse than the above-mentioned one where the coercer being the trusted helper since the voter is not aware of the problem here; however, like the previous threat, adding the original id to the Merkle tree would make it detectable by the voter.

5-A similar threat to the above, although less severe, is for an **eavesdropper** adversary who captures **the registration phase** keys (the case of leaked keys in [173/Table2]) then quickly vote with the true key before the voter. In fact, [173] mentions an axiom by [173/ ref. 21&22] that "*if the adversary has/knows/is everything that the voter has/knows/is, the voting system cannot distinguish the voter from the adversary and therefore cannot provide a mechanism to register the true voter's vote and not the adversary's*". However, the voter (if not coerced) can report to the authorities that the system rejected his/her vote for being already used when it is not; i.e., since sn depends solely on the registration phase values, the voter will be able to find it on BBB. Hence, this attack is detectable and could be resolved by following [173] in adding a time window between voting and tallying for such complaints.

6-As mentioned above, there is a **tradeoff between security and usability** in this design. In the original design, voters have to maintain an election identity "e", a zero-knowledge parameter "$param$", and at least 5 keys (master casting key pair, hybrid encryption key pair, ck^*, voting key pair, ck_{real}) in addition to any fake casting keys they will create. If EA generated all keys, a registering voter can still keep track of the numbers and check the correctness of the serial number "sn"; however, if voters left the details to the downloaded application and kept pressing the "OK" button without checking the numbers, this would make it easier for fake applications to deceive them.

8-Finally, there might be some **scalability** issues for country level elections where the paper implementation holds 2^{16} ballots ≈ 65,000 votes (the Merkle tree holding all vote hashes including fake ones). The author in the presentation [75/video] says the system handled over 100,000 votes since launched on 30 Nov 2024 with peak participation of 15,043 votes on 27 Dec 2024; for comparison basis, the Merkle tree of Zcash (2014), [170], was of 2^{64} (> 10^{18}) nodes. The scalability vs. trust issue was mentioned also in [166/sec. 3.3] who questions the public blockchain used in the actual implementation since the paper used a local Ethereum Ganache environment; if all used Ganache nodes and servers belong to the government, then the design loses most important public blockchain features. Recalling the quorum of trustees mentioned in [174]), we assume the standard threshold cryptography techniques between the government and other trusted third parties. In addition, like most existing e-voting systems that deploy blockchains, aggregated partial results and the final result are always posted on the public blockchain; recall the snapshots in Agora [20] and zk-rollups in Vocdoni [21]. For *zkVoting*, it will not add extra cost to post the Merkle root on Ethereum L1, and also periodical partial aggregations which helps voters to check not just their sn value is included but within the correct time interval they voted in; this makes it harder for fake applications to repeat the scenario in "4" for many voters using the same fake ballot.

5) Concluding Remarks

The idea of fake credentials remained in literature for so long, since 2005, before being used in actual elections. The given protocol should take the credit for an actual running implementation that scales up to 100,000 ballots if claims in [75/video] are true; we believe any performance efficiency issues could be improved following the original source of the nullifiers idea (Zcash [170]). We identified some threats and/or security vulnerabilities, some of which are common to all online voting systems; we believe most of them can be

solved by anchoring more pieces of data (a cryptographic fingerprint of IDs). Like all nullifiers and/or fake credential approaches, attending the whole registration phase (knowing all its keys) remains an undefeatable threat; when suggesting a time window between voting and tallying for an extra nullification phase, we should be aware that coercers can attend it as well.

B. Voatz

In this section we try to summarize and analyze the available material around Voatz with all the criticism that led to every US state taking a sperate independent decision about it; ex. [98].

1) Overview

Started in 2015 as an award-winning entrepreneur, conducted pilots in some USA states in 2018/2019 and expanded beyond many USA states to general elections in Mexico and 51 Ontario municipalities in Canada by now (https://voatz.com/blog/), Voatz is an online mobile voting system that is mostly used by remote voters. Exploring the official site (https://voatz.com/whitepapers/), earlier whitepapers [175,176] start by emphasizing the importance of providing solutions for persisting problems like less participation by remote voters and/or losing their votes by post office mail if arrived after deadline (recall footnote 6); as complained by [96,97] not very much technical and/or cryptographic details are included in them ([175] clearly states "*This paper is meant for a curious, non-technical reader*"). Mostly discussed voting steps are the authentication methods and the blockchain usage.

Voatz replaces the traditional public bulletin board (PBB) with a semi-private permissioned blockchain and public post-election tabulated ballots list [97/blockchain servers, 175,176]); a whitepaper by NASS (*National Association of Secretaries of State*) in 2025 [177] about audits via blockchain, cites [178] (2018) as the original peer-reviewed proposal of the blockchain usage idea. Another 2023 paper, [179/pages 29,30], is also featured in the official site for including a section about Voatz; the paper discusses the degree of decentralization in e-voting systems and considers the blockchain to be the only decentralized component in Voatz design, while the rest parts/phases take a centralized approach.

2) Main Voting Steps

The Voting steps, as illustrated in Fig.6 taken from [175], are:

1. **Voter Authentication** [175/KYV]: a voter is identified through a photo of ID/passport and a short interactive video to guarantee liveness, in addition to the smartphone/tablet authentication; i.e., each voter must have a 1-1 bond with a device and is required to re-authenticate whenever the application goes to the background. An anonymous voting ID is assigned to each voter which is a function of all the authentication parameters (including the device identifying information); we believe only Voatz, of all governmental level online voting systems, uses this authentication methodology.
2. **Voting through the blockchain** [175/Table2]: when the election starts, each jurisdiction assigns enough credit to eligible voters (to the anonymous voting ID) that is equal to number of choices. The voter choice for a certain candidate A is implemented as a transaction transferring credit from the voter to the candidate; in addition to cryptographic checks, the transaction is validated only (the voter account is debited and the candidate account is credited) if the choice is valid for this jurisdiction.
3. **Spoiling a Ballot**: a voter can spoil an old vote and vote again but must be through the same device.
4. **Ballot Receipt:** a voter receives a confirmation message when the ballot is validated and stored in the blockchain; the receipt contains the choices and the anonymized voter ID.
5. **Audits:** post-election audits are possible for voters, jurisdiction and independent viewers; this is through what could be called a PBB as in Fig.6 or a *Tabulated Ballot* (list) as called in Fig.7, where Fig.(7-a) is taken from [175] (2019) and Fig.(7-b) is taken from [177] (2025).

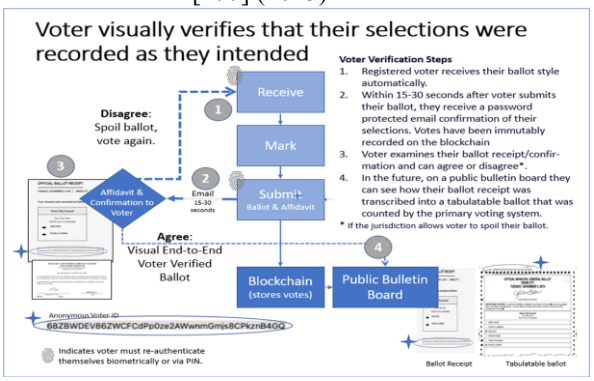


Fig. 6. Voter and verification steps in Voatz as shown in [175/Fig.2]

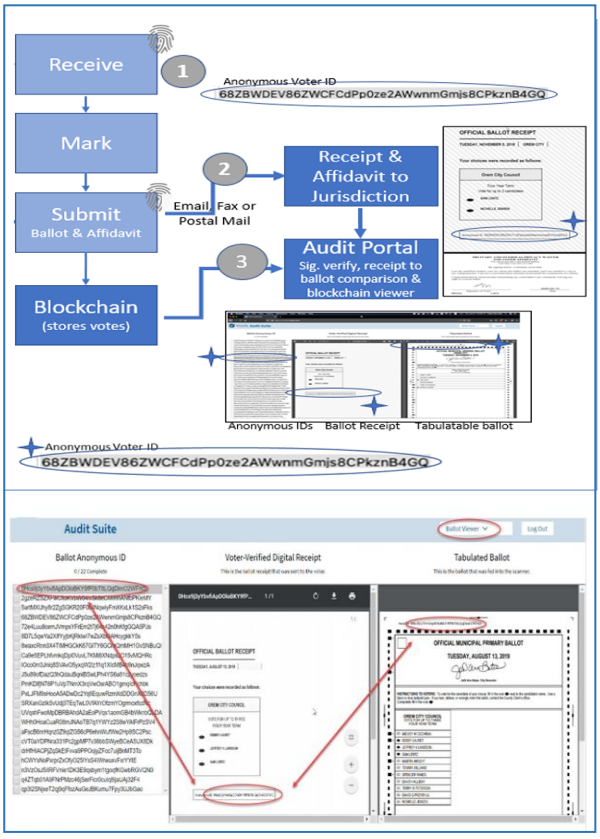


Fig. 7. Visual post-election audit procedure.

3) Known Details About Voatz Design

A main design pattern in Voatz is contracting with other companies for certain phases and/or services. Examples include:

1. Integrating (https://www.zimperium.com/) for **mobile threats defense** in USA. If the voter smartphone is found to be compromised, the device is prevented from voting; [180] shows some threat mitigations examples that happened during the pilots.
2. Deploying the **five-layer defense** strategy applied for remote voting in Mexico 2024 election using Amazon AWS (Amazon) [181].
3. The services provided by AWS for Mexico includes **image recognition and video analysis** for authentication, while (https://www.jumio.com/) was used in USA elections since 2019. According to [96/footnote 8] *Jumio* has disclosed then that it uses a third party, *Facetec*, to help in analyzing the video selfies.

As for **the blockchain design**, the main concern of this paper, released white papers [175,176] do not reveal much about the blockchain design except a FAQ table from which, among other resources like [96,97], we find the following:

1. Voatz uses a semi-private permissioned blockchain that consists of 32 decentralized storage servers (split equally between Amazon AWS and Microsoft's Azure in USA, while all Mexico servers are from AWS [181,182]; again, commercial companies). The blockchain servers run the IBM Hyperledger Fabric protocol on Linux and apply Practical Byzantine Fault Tolerance (PBFT) for consensus.
2. Anonymous ballot choices are directly stored in the blockchain in USA and all other elections run by Voatz; except for the Mexican elections, [182/sec. 2.5], where only the anonymous IDs are stored in it.
3. Since the blockchain is semi-private, Voatz still uses some form of a Public Bulletin Board (Fig.6&7) to store the post-election public auditable data; this public PBB is sometimes called *Tabulated Ballots list.*
4. Although not stated explicitly in white papers, we believe nullifiers might be used to spoil old votes in USA states. However, spoiling votes was not allowed on the Mexican elections and any attempt of repeated voting is detected as a replay attack; this aligns with the design choice of storing only anonymized IDs in the blockchain as there is a 1:1 correspondence with the number of stored ballots.
5. It is important to note that Voatz does not post any cryptographic checksums on any public permissionless blockchains.

For **cryptographic details**, we know the following:

1. According to [176/Appendix A: Communication & Network Security], all data transmitted over the network is encrypted using NIST approved AES-GCM 256 bits key (with RSA key agreement with SHA, SHA256, and SHA384); also, PKI (Public-private Key pair Infrastructure) is established between the voter device and the server. The PKI key generation and selection was one of the issues raised by MIT [96/sec. 4.1.1]; the researchers claimed that both sides generate 100 keys and the 57th key is always selected. However, Voatz replied [182/sec. 2.3] that this was in an earlier test version, but the version used in the election pilots makes a random selection.
2. The peer-reviewed paper about the Mexican elections [182/sec. 2] reveals the use of *Paillier* additively homomorphic encryption to encrypt ballots; i.e., encrypted ballots are multiplied then decrypted and no mix-nets are used. Also, a Non-Interactive Zero Knowledge Proof (*NIZKP*) is added to each vote online; ballots are signed with added randomness through *blind signature* that uses different prime numbers (p & q) than those used by *Paillier*; finally, *Fiat-Shamir secret sharing* was used to split the election authority key in a *3-out-of-5* threshold. However, we could not trace any document stating that those cryptographic details are applied to other elections conducted by Voatz whether in USA or Canada.

The rest available cryptographic details are better discussed one by one through the researchers' concerns.

4) Concerns raised by Researchers in the years 2019-2021

In addition to general concerns on online and/or blockchain-based voting, there were two research groups that analyzed Voatz in detail; one from MIT [96] and one from University of South Carolina [97]. The MIT group gained more publicity because they published their work in USENIX and because they were accused of hacking Voatz servers[22], while [97] remained as a preprint. This section discusses all concerns and/or vulnerabilities introduced by [96,97] along with Voatz reply to them [183]; since those debates took place in 2019/2020 and some replies concentrated on the pilot experiments only, we will try to discuss the general case (all eligible remote voters, not just a sampled group) and do the necessary follow up from the most updated site.

a) Certification

Voatz repeats in many places of [175,176] that it uses NIST (*National Institute of Science & Technology*) approved algorithms and applications. However, the authors in [97/certification] discussed that Voatz is not certified to the EAC (*Election Assistance Commission*), [184], standards and that EAC did not include standards for internet voting at that time. The authors did not deny though that Voatz seeks other security audit certificates from its affiliated partners (https://voatz.com/partners-and-affiliates/).

[22] We are not sure of whether Voatz proceeded with the accusations to report MIT researchers to the FBI or not, but the asserted fact is that there was a previous 2019 investigation with students from Michigan University who attempted to break into Voatz servers during 2018 pilots (as part of a course project activity on election cyber security); (https://www.michigandaily.com/section/news-briefs/university-michigan-students-implicated-potential-voting-app-hack, https://edition.cnn.com/2019/10/04/politics/fbi-voting-app-hack-investigation/index.html)

A significant audit was conducted by *Trail of Bits*, [185], in Mar 2020 to serve as a trusted third-party referee between Voatz and MIT researchers [96]. *Trail of Bits* reported 48 threats and classified 16 of them as severe; Voatz responded, [186], by resolving 20-22 of them while considering 17 as false positives and 9 as irrelevant. After the fixes, July 2020, Voatz successfully completed comprehensive system testing with a Federally certified VSTL (*Voting Systems Test Laboratory*), [187] as a phase1 requirement for EAC certification; the certificate says that Voatz "*meets the applicable requirements set forth for voting systems in the U.S. Election Assistance Commission (EAC) 2015 Voluntary Voting System Guidelines (VVSG), Version 1.1,* ***with clarifications or exceptions noted in Section 4.0***", unfortunately we found no trace of the complete report to read section 4 of it. However, the official site (https://voatz.com/audits/) shows 9 more independent security audits Voatz have conducted since then (mostly with its affiliated partners); the latest two (2025) are application penetration testing for Canada voters and an audit on the organization's ISMS (*Information Security Management System*) compliance (ISO/IEC 27001:2022).

b) Hiding Design Details & Control of Commercial Vendors

This was mentioned earlier under control of **blockchain server owners**; also, under transparency regarding the details of the exact **cryptography** used [96] and the exact fields stored on the **blockchain** and under what form.

Another concern that was highly raised by [96, 97/voter identification and authentication] is the methodology *Jumio* uses for **image and video analysis**; the exact machine learning and video analysis techniques used were hidden and the code is not open sourced. They questioned the threat of possible manipulation by the company; i.e., blocking certain voters on the claim that they failed the authentication check and/or accepting fake voters. Voatz, sufficed with replying, [183], that an additional human identification step is added to detect any identification errors whether malicious or not.

On the threat of **controlling the blockchain** by one of the vendor companies (Amazon & Microsoft), Voatz replied that it does not use a POW consensus to be subject to 51% attack; since the blockchain used is permissioned (PBFT consensus), companies cannot just add more nodes (servers) without permission. Although Voatz reply is right that 51% attack mistakenly mentioned by MIT in [96] is not possible, we remind that PBFT is resilient only with < 1/3 malicious nodes; i.e., their final ratio estimation of 1/3 the total power remains accurate though. Many of the questions raised by [97/blockchain servers] in 2019 are answered by now, except the concern about the possibility of a software bug in the consensus/operating and whether it would be repeated on half or all the servers. In fact, this complies with the threat we alarm about here and mentioned in footnote 13 and the section on controlling permissioned chains; PBFT can handle only <1/3 malicious nodes and each vendor controls 50% of the nodes. Had it hypothetically happened that one of those 2 vendors declared a certain bias in the US presidential elections, we doubt that Voatz will be able to convince cryptographers and blockchain experts that it is still Okey to let that company's servers constitute 50% of the blockchain nodes.

c) Altering/Discarding/Blocking the Vote

1. THROUGH MAN-IN-THE-MIDDLE ATTACK (MITM)

-Voatz stated in [176/Appendix A] that it uses *certificate pinning* to prevent/detect man-in-the-middle attacks; i.e., the application checks the server's cryptographic certificate against a local copy it has to make sure an adversary did not intercept in the middle and altered information sent between them.

-However, [96/sec. 4.1.1] claims that there is no authentication of the communication session keys (digital signatures) by the application, beyond the encapsulating TLS certificates which makes no protection against active MITM attacks. The researchers concluded that because it made it easier for them to direct the ballots to their own server (instead of Voatz servers) by simply replacing all required URLs in-app by their own and follow the protocol.

-Voatz on the other hand, [183/sec. 2.11/c)], denies those claims and describes them as "*blatantly incorrect*" asserting that public key authentication is performed as part of the initial voter onboarding. To explain the URLs easy replacement, Voatz accused the researchers of attempting to break into the official servers in Dec 2019 and denies their claim that Voatz servers were down during their experiment in Jan 2020.

2. THROUGH MALWARE OR FAKE APPLICATION

-As an outsider looking at the events to analyze it, if Voatz is right about the accusations and the researchers did not replace the URLs letting their experimental ballots go to the official Voatz servers, this means that the servers did not raise an alarm when receiving those fake ballots.

-If the researchers conducted their experiment in Dec 2019 (i.e., not in Jan 2020 as they claim) and the system backend interpreted the data they sent to the servers as a breaking attempt, this is not good enough for the voter side. Such a possibility means that if any malware on the voter device went undetected and tried to modify the voter choice using the same techniques used in [96] to modify the application original workflow, although the servers will be alarmed the voter is not.

-An analogous scenario is presented in [96/sec. 5.1.2/Stealth UI Modification Attack]; this was preceded by demonstrating how disabling the *Zimperium* malware defense SDK was possible in a few lines of code [96/Fig.7]. In their report, [185/p.20&60], Trail of Bits also pointed out that Zimperium can be circumvented (medium severity: TOB-VOATZ-029); Voatz on the other hand, [186], considered the threat a false positive.

-A third threat (in the general deployment if not in the pilot) is the possibility of the voter being deceived to download a fake malicious Voatz application. The fake app. Could show the voter similar screens to the original ones.

-A fourth threat introduced in [96/sec.] when discussing side channel attacks [96/sec. 5.3] is that biased ISPs (internet Service Providers) can block votes against their preference. If this is feasible, it is a threat even without side channel attacks; could prevent ballots from certain locations and/or IP

addresses if they have known dominating preferences for example.

In all these cases, the only defense [183/sec. 2.11/b)] is the voter's email receipt and its corresponding anonymous copy for officials and auditors; this is enough defense in a pilot experience but is it enough in the general case when applied to all remote voters. Practically, during general application, how many voters could be deceived by a fake email receipt or would not check it at all; how many would also proceed to search for their vote the final tabulated list. Furthermore, the third scenario of a fake application could be combined by a Bulletin Board *Clash attack* using one true vote for all voters with the same choice; the fake app can show them all the same anonymous ID Voatz provided the adversary for his/her only one counted vote.

Hence, unless there exist extra defenses that we and [96] missed, the system's integrity heavily depends on using a strong random sampling auditing mechanism to check errors and can benefit from a similar RLA (Risk Limiting Audit) approach as that in [188] (recall footnote 6).

d) Privacy Concerns

Although MIT researchers acknowledge [96/sec. 4.4] that "*Voatz employs a number of obfuscation techniques, leverages a third-party virus scanning service, and uses an on-device encrypted database to protect locally stored sensitive data*". They and [97] raised many privacy concerns and elaborated on their implications:

1. Privacy concerns stemmed from all the collected authentication data, binding each voter election ID to a fixed smartphone/tablet[23] device; [97] stressed on whether keeping the device ID after anonymization would leak some information about the identity of the voter, and the choice of the company handling biometric authentication. Both [96,97] questioned the guarantees that third parties will delete voters authentication data (copy of ID/passport, device pin, short video, …); MIT researchers also warned, [96/sec. 5.4] from collecting information about remote military deployments through voting. Voatz replied, [182], that all data transmissions are encrypted every step of the way and get deleted in a maximum of 24 hours.
2. The possibility of side channel attacks on the transmitted vote over the network (before anonymization) was discussed in [96/sec. 5.3] and featured as a main example in their presentation [96/video]. Voatz replied, [183/sec. 2.9] that the researchers used skewed flawed data to support their claim while different candidate choices are formed in the code to result in an almost balanced load. On the other hand, [96/sec.6] claims that the authors have alerted the U.S. Department of Homeland Security (DHS), and that "*they appeared to confirm the existence of the side channel vulnerability, and the PIN entropy issues*".
3. There were also concerns about being able to define the voter through the time of casting since a permissioned blockchain keeps the order and timing of transactions and there were not enough details on when (at any stage) mix-nets were used to shuffle the votes if used at all.

In addition to the defense replies mentioned above, Voatz simply reminded of the fact that remote voters already sign a waiver on their vote privacy rights when voting through regular mail. In their exact words [175/a note on absolute anonymity] "*However, given that every state needs to know who has voted (a requirement called "voter credit") and given that nearly every state requires the remote voter to waive their right to anonymity,* ***Voatz can make it extremely difficult, but not always impossible, for the election department staff to pair a voter's ballot with their identity***". However, some of those threats exceed the vote privacy to the voter personal data and could lead to discarding/blocking the vote if the adversary has the power to do so (ex. ISPs mentioned in the previous section.

5) Concluding Remarks on Voatz

We find most of the concerns raised about Voatz are not actually about its blockchain deployment but come from a common pattern of vulnerabilities any internet voting system might be susceptible to. At the end, it is a strategic decision for each country and/or state, which we recommend must be combined with strong auditing and sampling techniques (Risk Limiting Audits). However, we believe transparency is a major justified concern; we only learned some insight on the cryptographic details of the Mexican implementation through the peer-reviewed paper [182] and we still do not know if they are the same in the versions used for USA and Canada.

Regarding blockchain usage, when it is semi-private permissioned although the chain is controlled by those who have the permissioned access rights on them, concerns remain valid for the servers' owners (especially if they are commercial large-scale companies). On the other hand, the permissioned chain paradigm makes it possible to extend the permission rights beyond the governmental Election Authority (EA) and include independent observers; since PBFT consensus is resilient for up to f malicious nodes vs *2f+1* honest, one could imagine accepting *f+1* servers from independent observers for each *2f* government owned/controlled servers.

VI. Summary & Conclusions

In this paper we tried to gather and organize the existing material on the usage of blockchain technology in e-voting systems to be able to reach some conclusions.

We found two main approaches the first is to use a private or semi-private permissioned blockchain for storing encrypted ballots, **partially replacing the usual Bulletin Board**, and the second is to **post accumulated cryptographic checksums** of election data on one of famous public permissionless blockchains (i.e., Bitcoin or Ethereum L1). Some systems combine both approaches (ex. Agora,

[23]The authors in [97] doubted whether tablets are included, but the whitepaper [175] repeatedly writes smartphone/tablet which means that tablets are allowed.

Vocdoni); while some earlier research suggested posting all ballots on a public permissioned blockchain, this design is practical only for medium to small-scale elections. Table II summarizes example companies mentioned through the paper.

TABLE II. COMPANIES THAT DEPLOY BLOCKCHAINS IN VOTING

Company	***Ballot Storage***	***Cryptographic Checksums***	***Countries Using it***
Agora [20]	Customized Permissioned Chain (*SkipChain*)	Bitcoin (Catena scheme)	Sierra Leone Pilot (2018)
Opn.vote [22]	Inter Planetary File System (IPFS)	Meta data on Ethereum smart contract	Germany Pilot (2025)
SimpleProof [85,120]	NO	Bitcoin (Catena scheme)	-Guatemala 2nd round presidential elections (2023) by community -Some USA states (officially)
Vocdoni [21]	Ethereum L2 ZK-rollup	Ethereum L1	-The opposition in Belarus -Omnium Cultural organization in Barcelona
Voatz [175.176.182]	Permissioned Chain (Hyperledger Fabric)	NO	Some USA states, some Canada municipalities and Mexico 2024 elections
Zk-Voting [165]	Permissioned (Ethereum local Ganache chain)	NO	South Korea public institutions and local referenda

We collected most country level conducted pilots for blockchain-based e-voting to find out that they flooded the globe during the years 2016-2019 to settle down with a few real steady deployments in the years to follow. Posting cryptographic fingerprints on Bitcoin succeeded in Guatemala and some USA states, while blockchain bulletin boards are used in Russia, Mexico, and some USA states.

Looking through all discovered and/or literature discussed vulnerabilities from deploying blockchains in e-voting systems, we find that it is **a country's strategic decision that depends on many factors**. From election organizers point, it is the effect on election results they care about (read the quote from political *philanthropist Bradley Tusk* in [96/sec.6/conclusion]; for this argument, we recommend **accompanying such systems with the use of strong RLA technique** (ex. [188] ideas could be applied to Voatz too). Also, sometime the cost saving and increase in participation when combined with online voting significantly matter [176, footnote 15] to the state/country's decision; we anticipate Germany may allow internet voting (the proposed system is blockchain based) and adjust its law and/or regulations to allow it. In all cases, **we find posting cryptographic fingerprints to be always an added value with no added threats**; major countries who fear blockchain control may invest on having a ratio of the blockchain nodes and should assess a few large-scale commercial companies (the case of Voatz) as a higher severity threat.

In addition, this paper acknowledges **the added value of blockchain technology to research on cryptography** and Zero Knowledge Proofs (ZKPs) in specific; many e-voting systems benefited from *nullifiers* originally developed for Zcach and many more use (with or without blockchains) NIZKPs that were originally developed for a blockchain/cryptocurrency related purpose. The paper also point out to **new coercion and vote buying innovative techniques that became possible through the existence of blockchains** public immutable ledger and/or smart contracts online coding capabilities; it is quite feasible to conduct online vote auctions, to automatically vote with the adversary's choice in some systems, and for the coercers to force a coerced voter to post some intermediate revealing values on a public blockchain. Again, all those threats do not depend on deploying blockchains in the used e-voting system.

Finally, we detailed two case study systems in this paper. The first is **zk-voting with nullifiable commitments** in South Korea and compared it with another system that uses nullifiers without blockchains. Then, we got into the details of **Voatz** with all the debates and criticism that happened around it in USA (2019-2021) and discussed its 2024 Mexican modified system as well.

We hope this paper provides added value for policy makers on their decisions on whether to use blockchains on their e-voting systems after reading all the relevant facts; also, to become aware of possible new attacks even if they did not use blockchains. As for academic researchers, whether on e-voting or blockchain areas, we hope the paper provided a thorough survey and accurate analysis/concluding remarks that constitute a starting point incentivizing new research.

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The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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